

Other solutions in smart city administration

Cloud service providers like AWS and Azure provide blockchain network service components like Hyperledger Fabric using custom cloud service implementation and integration with other cloud service components.

For example, Azure provides cloud service integration for Hyperledger solutions (as shown in Figure 2) using:

- VPC peering for peer node integration
- Interface development using Service Fabric
- Resource group management for handling Hyper Fabric
- Data store for ledger management services
- App services for rule implementation

This architecture can be implemented using AWS based architecture by using VPC peering, peer node services and Fabric components, as shown in Figure 3.

Future implementations in smart city administration using blockchain

Blockchain could be a great ally in the development of smart cities. Indeed, the need for the use of this technology is closely aligned with the needs of smart cities, which are:

- Multiple agents sharing information (society is hyperconnected and will become ever more so; the number of devices per person will also grow).
- Constant updation of data.
- The need to verify that shared data—and updates thereof—are valid.
- Simplification of management, which gets complicated due to the existence of intermediaries.
- Fast-paced, agile connections and participants who interact with one another.

Given this cocktail, the added value of blockchain lies in the possibility of sharing only the information that a participant wishes

to know, while keeping everything else encrypted and inaccessible. Cryptography can be used to prevent manipulation, modification of information and privacy violations. It seems logical that the ‘de-intermediated’ agent—the city government—will try to resist the implementation of blockchain. In the world of *Blockchain4Cities*, however, business models are under no threat. On the contrary, public administrators will have no problem relinquishing their intermediary role so long as all the other roles having to do with control, security, privacy, efficiency, transparency, etc, remain intact.

Cities such as Santiago, Toronto, Tel Aviv, Oslo, Milan, London and Stockholm are already developing blockchain based projects. Dubai has become the world capital of blockchain development by launching the Dubai Blockchain Platform and the Dubai Blockchain Policy.

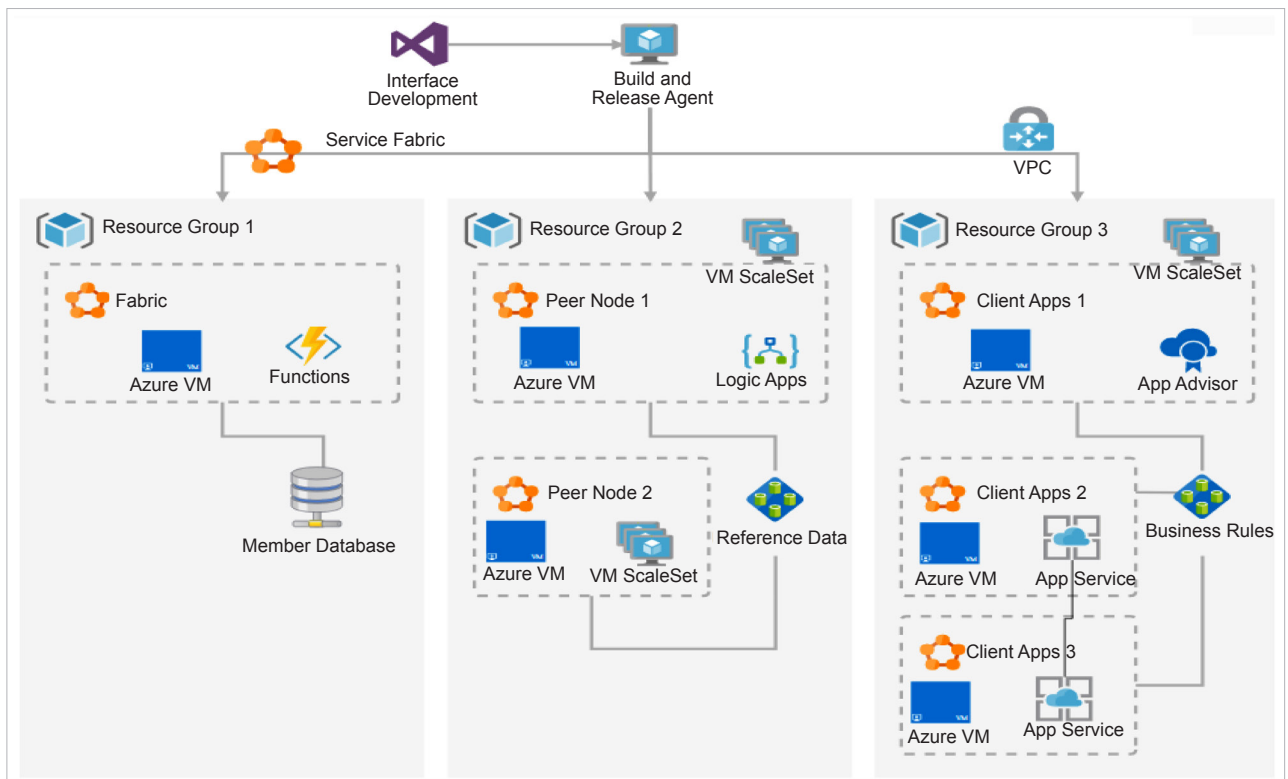


Figure 3: Azure implementation of blockchain architecture